

Dissolved Oxygen

HD-200

(Four-Wire Analyzer)



HD-300

(Two-Wire Transmitter)



HD-200 Specifications

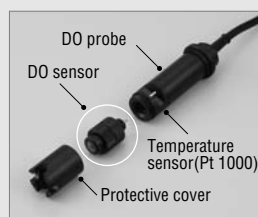
Measuring method	Membrane type polarography method
Measuring range	Dissolved oxygen concentration: 0 to 20 mg/L Resolution: 0.01 mg/L Saturation ratio: 0 to 200% Resolution: 1% Temperature: 0 to 50°C Resolution: 0.1°C
Repeatability	Dissolved oxygen concentration: $\pm 0.5\%$ full-scale or less, Temperature: $\pm 0.5^\circ\text{C}$ (for equivalent input)
Linearity	Dissolved oxygen concentration: $\pm 0.5\%$ full-scale or less, Temperature: $\pm 0.5^\circ\text{C}$ (for equivalent input)
Transmission output	Two points 4 to 20 mA DC Input/output insulated type Maximum load resistance 900 Ω
Contact output	Three points Non-voltage contact output Relay contact, SPDT(1c) Contact function R1, R2: Selectable from upper limit alarm, lower limit alarm, ON/OFF control, transmission output hold operation, cleaning output FAIL: Malfunction alarm Control operation Control range: 0.02 to 1.00 mg/L (± 0.01 to ± 0.50 mg/L)
Contact input	One point Contact format: Open collector non-voltage a contact Contact function: Cleaning operation external input
Communication function	RS-485 Two wire system, Input/output insulated type (not insulated for transmission output)
Temperature compensation range	0 to 50°C
Ambient temperature	-20 to 55°C
Temperature measurement element	Pt 1000 (0°C) (Built into DO-1100 probe)
Calibration method	Atmospheric calibration or saturated liquid calibration
Self-diagnosis function	Calibration error, Sensor diagnosis error, Analyzer malfunction
Power supply	90 to 264 V AC 50/60 Hz Power consumption 15 W or less
Construction	Outdoor installation type: IP65 protection level Mounting method: 50A pole or wall mounted Case: Aluminum alloy Mount fitting/hood: SUS304
Compatible sensor	Sensor: 5505, 5510 Probe: DO-1100
Weight	Approx. 4.5 kg
Equipment protected	CE marking, FCC rules

HD-300 Specifications

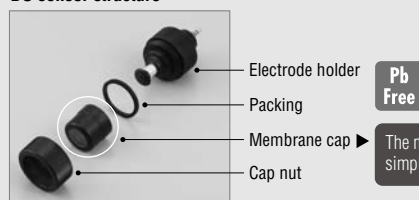
Measuring method	Membrane type polarography method
Measuring range	Dissolved oxygen: 0 to 20 mg/L Resolution: 0.01 mg/L Saturation ratio: 0 to 200% Resolution: 1% Temperature: 0 to 50°C Resolution: 0.1°C
Repeatability	Dissolved oxygen concentration: $\pm 0.5\%$ full-scale or less, Temperature: $\pm 0.5^\circ\text{C}$ (for equivalent input)
Linearity	Dissolved oxygen concentration: $\pm 0.5\%$ full-scale or less, Temperature: $\pm 0.5^\circ\text{C}$ (for equivalent input)
Transmission output	4 to 20 mA DC Input/output insulated type Maximum load resistance 600 Ω
Contact input	One point Contact format: Non-voltage a contact Contact function: Transmission output is held for closed contact input.
Temperature compensation range	0 to 50°C
Ambient temperature	-20 to 60°C
Temperature measurement element	Pt 1000 (0°C) (Built into DO-1100 probe)
Calibration method	Atmospheric calibration or saturated liquid calibration
Self-diagnosis function	Calibration error, Sensor diagnosis error, Transmitter malfunction
Power supply	24 V DC (operational voltage range: 21 V to 32 V DC) Power consumption 0.6 W or less
Construction	Outdoor installation type: IP65 protection level Mounting method: 50A pole or wall mounted Case: Aluminum alloy Mount fitting/hood: SUS304
Compatible sensor	Sensor: 5505, 5510 Probe: DO-1100
Weight	Approx. 4 kg
Equipment protected	CE marking, FCC rules

DO sensing system for HD-200 / HD-300

Combines sensor replacement and internal solution, membrane replacement during parts renewal



DO sensor structure



The membrane remains fixed to the diaphragm cap, simplifying the replacement work during tip renewal.

Specifications

Product name	5505	5510
Measuring method	Membrane type polarography	
Construction	Sensor replacement and membrane internal solution replacement system	
Material	Membrane thickness	50 μm 100 μm
	Membrane material	PFA
	Electrode material	C-Ag
	Solution contact material	PPO, PFA, EPDM
	Internal solution	KCl (neutral)
Performance	Response speed (90% response)	120 sec. or less 240 sec. or less
	Repeatability	± 0.1 mg/L
Measurement solution conditions	Temperature	0 to 50°C
	Pressure	0 to 0.5 MPa
	Flow velocity	20 cm/sec. or more 10 cm/sec. or more
Combined probe	DO-1100	

Caution: The selection of equipment will differ based on various conditions, including the installation site, usage environment, measurement samples, and any special characteristics. Contact your sales representative for details.